Assignment 17.1

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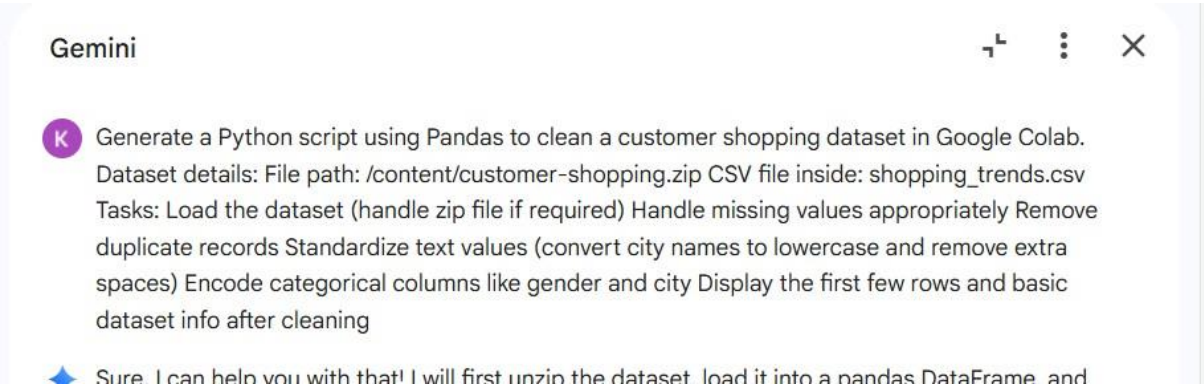
B27

Task 1 – Customer Data Cleaning Prompt:

Generate a Python script using Pandas to clean a customer shopping dataset in Google Colab. Dataset details: File path: /content/customer-shopping.zip CSV file inside: shopping_trends.csv Tasks: Load the dataset (handle zip file if required) Handle missing values appropriately Remove duplicate records Standardize text values (convert city names to lowercase and remove extra spaces) Encode categorical columns like gender and city Display the first few rows and basic dataset info after cleaning



CODE:

```python
import zipfile
import pandas as pd
import os

# Define the path to the zip file and the output directory
zip_file_path = '/content/customer-shopping.zip'
extract_dir = '/content/'

# Create the extraction directory if it doesn't exist
os.makedirs(extract_dir, exist_ok=True)

# Extract the zip file
with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    zip_ref.extractall(extract_dir)

print(f"'{os.listdir(extract_dir)[0]}' extracted successfully to {extract_dir}")

# Load the dataset
df = pd.read_csv(os.path.join(extract_dir, 'shopping_trends.csv'))

print("Original DataFrame info:")
df.info()
display(df.head())
```

```python
print("Missing values before handling:")
display(df.isnull().sum())

# For simplicity, let's fill numerical columns with their median and categorical with their mode
# Identify numerical and categorical columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
categorical_cols = df.select_dtypes(include=['object']).columns

# Fill numerical missing values with the median
for col in numerical_cols:
    if df[col].isnull().any():
        df[col].fillna(df[col].median(), inplace=True)

# Fill categorical missing values with the mode
for col in categorical_cols:
    if df[col].isnull().any():
        df[col].fillna(df[col].mode()[0], inplace=True)

print("\nMissing values after handling:")
display(df.isnull().sum())
```

Next, we'll remove any duplicate records to ensure data integrity.

```python
initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
cleaned_rows = df.shape[0]
print(f"Number of duplicate records removed: {initial_rows - cleaned_rows}")
print(f"DataFrame shape after removing duplicates: {df.shape}")
```

+ Code + Text

[4]

```python
string_cols = df.select_dtypes(include=['object']).columns

for col in string_cols:
    # Convert to string to handle potential non-string types, then to lowercase and strip spaces
    df[col] = df[col].astype(str).str.lower().str.strip()

print("Text values standardized. Here are the first few rows with standardized text:")
display(df.head())
```

[7]
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```python
print(f"Unique Genders: {df['Gender'].nunique()}")
print(f"Unique Locations: {df['Location'].nunique()}")

# One-hot encode 'Gender' and 'Location'
df = pd.get_dummies(df, columns=['Gender', 'Location'], drop_first=True)

print("\nDataFrame after encoding categorical columns:")
df.info()
display(df.head())
```

[6]

```python
print("Final Cleaned DataFrame Info:")
df.info()
print("\nFirst 5 rows of the cleaned DataFrame:")
display(df.head())
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Customer ID               3900 non-null   int64
 1   Age                       3900 non-null   int64
 2   Gender                    3900 non-null   object
 3   Item Purchased            3900 non-null   object
 4   Category                  3900 non-null   object
 5   Purchase Amount (USD)     3900 non-null   int64
 6   Location                  3900 non-null   object
 7   Size                      3900 non-null   object
 8   Color                     3900 non-null   object
 9   Season                    3900 non-null   object
 10  Review Rating             3900 non-null   float64
 11  Subscription Status       3900 non-null   object
 12  Payment Method            3900 non-null   object
 13  Shipping Type             3900 non-null   object
 14  Discount Applied          3900 non-null   object
 15  Promo Code Used           3900 non-null   object
 16  Previous Purchases        3900 non-null   int64
 17  Preferred Payment Method  3900 non-null   object
 18  Frequency of Purchases    3900 non-null   object
dtypes: float64(1), int64(4), object(14)
memory usage: 579.0+ KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Credit Card	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Bank Transfer	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Cash	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	PayPal	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Cash	Free Shipping	Yes	Yes	31

```
display(df.isnull().sum())
```

```
Missing values before handling:
```

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Payment Method	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Preferred Payment Method	0
Frequency of Purchases	0

dtype: int64

```
Missing values after handling:
```

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0

```
Number of duplicate records removed: 0
DataFrame shape after removing duplicates: (3900, 19)
```

+ Code + Text

Now, we will standardize text values, specifically converting city names to lowercase and removing

```
Number of duplicate records removed: 0
DataFrame shape after removing duplicates: (3900, 19)
```

Now, we will standardize text values, specifically converting city names to lowerc

Text values standardized. Here are the first few rows with standardized text:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Frequency of Purchases
0	1	55	male	blouse	clothing	53	kentucky	l	gray	winter	3.1	yes	credit card	express	yes	yes	14	venmo	fortnightly
1	2	19	male	sweater	clothing	64	maine	l	maroon	winter	3.1	yes	bank transfer	express	yes	yes	2	cash	fortnightly
2	3	50	male	jeans	clothing	73	massachusetts	s	maroon	spring	3.1	yes	cash	free shipping	yes	yes	23	credit card	weekly
3	4	21	male	sandals	footwear	90	rhode island	m	maroon	spring	3.5	yes	paypal	next day air	yes	yes	49	paypal	weekly
4	5	45	male	blouse	clothing	49	oregon	m	turquoise	spring	2.7	yes	cash	free shipping	yes	yes	31	paypal	annually

```
Unique Genders: 2
Unique Locations: 50

DataFrame after encoding categorical columns:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 67 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Customer ID               3900 non-null   int64
 1   Age                       3900 non-null   int64
 2   Item Purchased            3900 non-null   object
 3   Category                  3900 non-null   object
 4   Purchase Amount (USD)     3900 non-null   int64
 5   Size                      3900 non-null   object
 6   Color                     3900 non-null   object
 7   Season                    3900 non-null   object
 8   Review Rating             3900 non-null   float64
 9   Subscription Status       3900 non-null   object
 10  Payment Method            3900 non-null   object
 11  Shipping Type             3900 non-null   object
 12  Discount Applied          3900 non-null   object
 13  Promo Code Used           3900 non-null   object
 14  Previous Purchases        3900 non-null   int64
 15  Preferred Payment Method  3900 non-null   object
 16  Frequency of Purchases    3900 non-null   object
 17  Gender_male               3900 non-null   bool
 18  Location_alaska           3900 non-null   bool
 19  Location_arizona          3900 non-null   bool
 20  Location_arkansas         3900 non-null   bool
 21  Location_california       3900 non-null   bool
 22  Location_colorado         3900 non-null   bool
 23  Location_connecticut      3900 non-null   bool
 24  Location_delaware         3900 non-null   bool
 25  Location_florida          3900 non-null   bool
 26  Location_georgia          3900 non-null   bool
 27  Location_hawaii           3900 non-null   bool
 28  Location_idaho            3900 non-null   bool
 29  Location_illinois         3900 non-null   bool
 30  Location_indiana          3900 non-null   bool
 31  Location_iowa             3900 non-null   bool
 32  Location_kansas           3900 non-null   bool
 33  Location_kentucky         3900 non-null   bool
 34  Location_louisiana        3900 non-null   bool
 35  Location_maine            3900 non-null   bool
```

```
memory usage: 708.5+ KB
```

	Customer ID	Age	Item Purchased	Category	Purchase Amount (USD)	Size	Color	Season	Review Rating	Subscription Status	...	Location_south dakota	Location_tennessee	Location_texas	Location_utah	Location_vermont	Location_virginia	Location_washington
0	1	55	blouse	clothing	53	l	gray	winter	3.1	yes	...	False	False	False	False	False	False	False
1	2	19	sweater	clothing	64	l	maroon	winter	3.1	yes	...	False	False	False	False	False	False	False
2	3	50	jeans	clothing	73	s	maroon	spring	3.1	yes	...	False	False	False	False	False	False	False
3	4	21	sandals	footwear	90	m	maroon	spring	3.5	yes	...	False	False	False	False	False	False	False
4	5	45	blouse	clothing	49	m	turquoise	spring	2.7	yes	...	False	False	False	False	False	False	False

5 rows × 67 columns

```
display(df.head())

Final Cleaned DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3900 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Payment Method          3900 non-null   object
 13  Shipping Type           3900 non-null   object
 14  Discount Applied        3900 non-null   object
 15  Promo Code Used         3900 non-null   object
 16  Previous Purchases      3900 non-null   int64
 17  Preferred Payment Method 3900 non-null  object
 18  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(14)
memory usage: 579.0+ KB

First 5 rows of the cleaned DataFrame:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Frequency of Purchases
0	1	55	male	blouse	clothing	53	kentucky	l	gray	winter	3.1	yes	credit card	express	yes	yes	14	venmo	fortnightly
1	2	19	male	sweater	clothing	64	maine	l	maroon	winter	3.1	yes	bank transfer	express	yes	yes	2	cash	fortnightly
2	3	50	male	jeans	clothing	73	massachusetts	s	maroon	spring	3.1	yes	cash	free shipping	yes	yes	23	credit card	weekly
3	4	21	male	sandals	footwear	90	rhode island	m	maroon	spring	3.5	yes	paypal	next day air	yes	yes	49	paypal	weekly
4	5	45	male	blouse	clothing	49	oregon	m	turquoise	spring	2.7	yes	cash	free shipping	yes	yes	31	paypal	annually

EXPLANATION:

- Clean missing data: Fix or fill empty values in important columns so the dataset becomes usable and consistent.
- Remove duplicates: Delete repeated rows to avoid incorrect analysis results.
- Standardize text: Make values like city names uniform (e.g., lowercase, no extra spaces) so they don't get treated as different entries.
- Prepare for ML: Convert categorical data (like gender and city) into numeric form so it can be used for analysis or machine learning models.

Task 2 – Sales Data Preprocessing Prompt:

Generate a Python script using Pandas to preprocess a sales dataset in Google Colab. Dataset details: file path/content/salesdataset.zip : File name: sales_data.csv Tasks: Load the dataset Convert date column into datetime format Create new features: month, year, and day of week from the date

column Normalize the sales amount column Detect and handle outliers in the dataset Display the first few rows after preprocessing Requirements: Keep the code simple and clear Add basic comments for understanding Final output should be a processed dataset ready for analysis

K Generate a Python script using Pandas to preprocess a sales dataset in Google Colab. Dataset details: * file path/content/salesdataset.zip : File name: sales_data.csv Tasks: * Load the dataset * Convert date column into datetime format * Create new features: month, year, and day of week from the date column * Normalize the sales amount column * Detect and handle outliers in the dataset * Display the first few rows after preprocessing Requirements: * Keep the code simple and clear * Add basic comments for understanding * Final output should be a processed dataset ready for analysis

CODE:

```python
import zipfile
import pandas as pd
import numpy as np
import os
from sklearn.preprocessing import MinMaxScaler

# Define the path to the zip file and the output directory
zip_file_path = '/content/salesdataset.zip'
extract_dir = '/content/'
csv_file_name = 'sales_data.csv'
full_csv_path = os.path.join(extract_dir, csv_file_name)

# Create the extraction directory if it doesn't exist
os.makedirs(extract_dir, exist_ok=True)

# Extract the zip file
try:
    with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
        zip_ref.extractall(extract_dir)
    print(f"'{zip_file_path}' extracted successfully to {extract_dir}")
    # Load the dataset directly after successful extraction
    sales_df = pd.read_csv(full_csv_path)
    print(f"'{csv_file_name}' loaded successfully.")
except FileNotFoundError:
    print(f"Error: Zip file '{zip_file_path}' not found.")
    # Create a dummy DataFrame if the zip file is not found
    print("Creating a dummy DataFrame for demonstration.")
    data = {
        'Date': pd.to_datetime(pd.date_range(start='2023-01-01', periods=100)),
        'SalesAmount': np.random.randint(100, 1000, size=100) + np.random.normal(0, 50, size=100),
        'ProductID': np.random.choice(['A', 'B', 'C', 'D'], size=100),
        'Region': np.random.choice(['East', 'West', 'North', 'South'], size=100)
    }
    sales_df = pd.DataFrame(data)
    # Introduce some outliers for demonstration
    sales_df.loc[5, 'SalesAmount'] = 5000
    sales_df.loc[15, 'SalesAmount'] = 6000
    sales_df.loc[25, 'SalesAmount'] = 50 # Example of a low outlier
    # Ensure 'Date' is datetime if dummy data is used
    sales_df['Date'] = pd.to_datetime(sales_df['Date'])
except Exception as e:
    print(f"An unexpected error occurred: {e}")
    print("Attempting to create a dummy DataFrame to proceed.")
    data = {
        'Date': pd.to_datetime(pd.date_range(start='2023-01-01', periods=100)),
        'SalesAmount': np.random.randint(100, 1000, size=100) + np.random.normal(0, 50, size=100),
        'ProductID': np.random.choice(['A', 'B', 'C', 'D'], size=100),
        'Region': np.random.choice(['East', 'West', 'North', 'South'], size=100)
    }
    sales_df = pd.DataFrame(data)
    sales_df.loc[5, 'SalesAmount'] = 5000
    sales_df.loc[15, 'SalesAmount'] = 6000
    sales_df.loc[25, 'SalesAmount'] = 50
    sales_df['Date'] = pd.to_datetime(sales_df['Date'])

print("\nOriginal DataFrame info:")
sales_df.info()
print("\nFirst 5 rows of the original DataFrame:")
display(sales_df.head())
```

```python
# Ensure 'Sale_Date' column is in datetime format
sales_df['Sale_Date'] = pd.to_datetime(sales_df['Sale_Date'])

# Create new features: Month, Year, Day of Week
sales_df['Month'] = sales_df['Sale_Date'].dt.month
sales_df['Year'] = sales_df['Sale_Date'].dt.year
sales_df['DayOfWeek'] = sales_df['Sale_Date'].dt.dayofweek # Monday=0, Sunday=6

print("DataFrame after adding date features:")
display(sales_df.head())
```

[19]
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```python
# Initialize MinMaxScaler
scaler = MinMaxScaler()

# Reshape the 'Sales_Amount' column for scaling (MinMaxScaler expects a 2D array)
sales_df['Sales_Amount_Normalized'] = scaler.fit_transform(sales_df[['Sales_Amount']])

print("DataFrame after normalizing 'Sales_Amount':")
display(sales_df.head())
```

```python
# Calculate IQR for 'Sales_Amount'
Q1 = sales_df['Sales_Amount'].quantile(0.25)
Q3 = sales_df['Sales_Amount'].quantile(0.75)
IQR = Q3 - Q1

# Define outlier bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Identify outliers
outliers_count = sales_df[(sales_df['Sales_Amount'] < lower_bound) | (sales_df['Sales_Amount'] > upper_bound)].shape[0]
print(f"Number of outliers detected in 'Sales_Amount': {outliers_count}")

# Handle outliers by capping (replacing values outside bounds with the bounds themselves)
sales_df['Sales_Amount_Cleaned'] = sales_df['Sales_Amount'].clip(lower=lower_bound, upper=upper_bound)

print("\nDataFrame after handling outliers in 'Sales_Amount':")
display(sales_df.head())
```

```python
print("Final Processed DataFrame Info:")
sales_df.info()
print("\nFirst 5 rows of the processed DataFrame:")
display(sales_df.head())
```

OUTPUT:

```
'/content/salesdataset.zip' extracted successfully to /content/
'sales_data.csv' loaded successfully.

Original DataFrame info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Product_ID            1000 non-null   int64
 1   Sale_Date             1000 non-null   object
 2   Sales_Rep             1000 non-null   object
 3   Region                1000 non-null   object
 4   Sales_Amount          1000 non-null   float64
 5   Quantity_Sold         1000 non-null   int64
 6   Product_Category      1000 non-null   object
 7   Unit_Cost             1000 non-null   float64
 8   Unit_Price            1000 non-null   float64
 9   Customer_Type         1000 non-null   object
 10  Discount              1000 non-null   float64
 11  Payment_Method        1000 non-null   object
 12  Sales_Channel         1000 non-null   object
 13  Region_and_Sales_Rep  1000 non-null   object
dtypes: float64(4), int64(2), object(8)
memory usage: 109.5+ KB

First 5 rows of the original DataFrame:
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_I
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-E
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-E
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-Da
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-E
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Cha

DataFrame after adding date features:

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	Month	Year	DayOfWeek
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4

DataFrame after normalizing 'Sales_Amount':

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	Month	Year	DayOfWeek	Sales
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4	
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4	
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3	
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3	
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4	

Number of outliers detected in 'Sales_Amount': 0

DataFrame after handling outliers in 'Sales_Amount':

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	Month	Year	DayOfWeek	Sales_A
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4	
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4	
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3	
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3	
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4	

```
display(sales_df.head())

Final Processed DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 19 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Product_ID              1000 non-null   int64
 1   Sale_Date               1000 non-null   datetime64[ns]
 2   Sales_Rep               1000 non-null   object
 3   Region                  1000 non-null   object
 4   Sales_Amount            1000 non-null   float64
 5   Quantity_Sold           1000 non-null   int64
 6   Product_Category        1000 non-null   object
 7   Unit_Cost               1000 non-null   float64
 8   Unit_Price              1000 non-null   float64
 9   Customer_Type           1000 non-null   object
 10  Discount                1000 non-null   float64
 11  Payment_Method          1000 non-null   object
 12  Sales_Channel           1000 non-null   object
 13  Region_and_Sales_Rep    1000 non-null   object
 14  Month                   1000 non-null   int32
 15  Year                    1000 non-null   int32
 16  DayOfWeek               1000 non-null   int32
 17  Sales_Amount_Normalized 1000 non-null   float64
 18  Sales_Amount_Cleaned    1000 non-null   float64
dtypes: datetime64[ns](1), float64(6), int32(3), int64(2), object(7)
memory usage: 136.8+ KB

First 5 rows of the processed DataFrame:
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	Month	Year	DayOfWeek	Sales_Amount_Normalized	Sales_Amount_Cleaned
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4	0.500960	5053.97
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4	0.433202	4384.02
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3	0.458201	4631.23
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3	0.209105	2167.94
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4	0.369108	3750.20

EXPLANATION:

- Prepare date information: Convert the date column into proper datetime format so it can be used for analysis.
- Extract time-based features: Create new columns like month, year, and day of the week to understand sales trends over time.
- Normalize sales values: Scale the sales amount to a consistent range to improve analysis and model performance.
- Handle outliers: Identify and treat extreme values to prevent them from affecting the accuracy of insights.

Task 3 – Healthcare Data Preparation PROMPT:

datasetpath:/content/HealtCare_Risk_factor.zip Generate Python code to clean and preprocess a healthcare dataset. Steps: - Fill missing values in blood_pressure and cholesterol - Encode categorical variables - Scale numerical features - Split data into train and test sets

Gemini × app.log

K datasetpath:/content/HealtCare_Risk_factor.zip Generate Python code to clean and preprocess a healthcare dataset. Steps: - Fill missing values in blood_pressure and cholesterol - Encode categorical variables - Scale numerical features - Split data into train and test sets

CODE:

Unzipping the Dataset

First, I'll unzip the `HealtCare_Risk_factor.zip` file to extract the dataset. I'll assume the extracted file is a CSV and name it `HealtCare_Risk_factor.csv`.

```
!unzip -o /content/HealtCare_Risk_factor.zip -d /content/

# List the contents to verify the extracted file name
!ls -l /content/
```

```
Archive:  /content/HealtCare_Risk_factor.zip
  inflating: /content/dirty_v3_path.csv
total 4620
-rw-r--r-- 1 root root 3379281 Sep 27  2025 dirty_v3_path.csv
-rw-r--r-- 1 root root 1342385 Apr  1 16:03 HealtCare_Risk_factor.zip
drwxr-xr-x 1 root root    4096 Mar 30 13:29 sample_data
```

Load the Dataset

Now, I'll load the extracted CSV file into a pandas DataFrame and display its first few rows and information to understand its structure and check for initial data types and missing values.

```
import pandas as pd

# The extracted CSV file is named 'dirty_v3_path.csv'
try:
    df = pd.read_csv('/content/dirty_v3_path.csv')
except FileNotFoundError:
    print("Error: 'dirty_v3_path.csv' not found. Please check the extracted file name.")
    exit()

print("First 5 rows of the dataset:")
display(df.head())

print("\nDataset information:")
display(df.info())
```

Handle Missing Values

I'll fill missing values in the `blood_pressure` and `cholesterol` columns. For numerical columns, the median is often a robust choice to avoid distortion from outliers. I'll also check and fill any other missing values if they exist, possibly using the median for numerical columns and the mode for categorical columns.

[6]

```
from sklearn.impute import SimpleImputer
import numpy as np

# Impute 'Blood Pressure' and 'Cholesterol' if they exist and have missing values
for col in ['Blood Pressure', 'Cholesterol']:
    if col in df.columns:
        if df[col].isnull().any():
            median_val = df[col].median()
            df[col] = df[col].fillna(median_val)
            print(f"Missing values in '{col}' filled with median: {median_val}")
        else:
            print(f"No missing values in '{col}'.")
    else:
        print(f"Column '{col}' not found in the dataset.")

# Identify other columns with missing values and impute them
missing_cols = df.columns[df.isnull().any()].tolist()
if missing_cols:
    print("\nImputing other missing values:")
    for col in missing_cols:
        if df[col].dtype == 'object': # Categorical
            mode_val = df[col].mode()[0]
            df[col] = df[col].fillna(mode_val)
            print(f"  Missing values in categorical column '{col}' filled with mode: {mode_val}")
        elif df[col].dtype in ['int64', 'float64']: # Numerical
            median_val = df[col].median()
            df[col] = df[col].fillna(median_val)
            print(f"  Missing values in numerical column '{col}' filled with median: {median_val}")
else:
    print("\nNo other missing values found in the dataset.")

print("\nMissing values after imputation:")
display(df.isnull().sum())
```

Separate Target Variable (y) and Features (X)

Before encoding categorical variables and scaling numerical features, I'll separate the target variable, 'Medical Condition', from the rest of the features. This ensures that the target column is not modified during the feature preprocessing steps.

[7]

```
# Define the target column name
target_column_name = 'Medical Condition'

if target_column_name in df.columns:
    y = df[target_column_name]
    X = df.drop(columns=[target_column_name])
    print(f"Target variable '{target_column_name}' separated successfully.")
    print(f"X shape: {X.shape}")
    print(f"y shape: {y.shape}")
else:
    print(f"Error: Target column '{target_column_name}' not found after missing value imputation. Please check the DataFrame content and preceding steps.")
    exit()
```

Encode Categorical Variables

I'll convert categorical features into a numerical format using One-Hot Encoding. This is necessary for most machine learning algorithms. I'll identify columns with 'object' dtype as categorical.

```
from sklearn.preprocessing import OneHotEncoder

# Identify categorical columns in X
categorical_cols = X.select_dtypes(include=['object']).columns

if not categorical_cols.empty:
    print(f"Categorical columns identified in features (X): {list(categorical_cols)}")
    # Initialize OneHotEncoder
    encoder = OneHotEncoder(sparse_output=False, handle_unknown='ignore')

    # Apply one-hot encoding to X's categorical columns
    encoded_features = encoder.fit_transform(X[categorical_cols])

    # Create a DataFrame for encoded features
    encoded_X_df = pd.DataFrame(encoded_features, columns=encoder.get_feature_names_out(categorical_cols), index=X.index)

    # Drop original categorical columns from X and concatenate encoded ones
    X = X.drop(columns=categorical_cols)
    X = pd.concat([X, encoded_X_df], axis=1)
    print("Categorical columns in X encoded successfully.")
else:
    print("No categorical columns found in features (X) for encoding.")

print("\nFeatures (X) after one-hot encoding:")
display(X.head())
```

Scale Numerical Features

Next, I'll scale numerical features using `StandardScaler`. This standardizes the data by removing the mean and scaling to unit variance, which can improve the performance of many machine learning models.

```
from sklearn.preprocessing import StandardScaler

# Identify numerical columns in X
numerical_cols = X.select_dtypes(include=np.number).columns.tolist()

# Exclude any potential ID columns if they exist (e.g., 'id', 'patient_id')
# This is a heuristic; adjust if specific ID columns are known.
id_cols = [col for col in numerical_cols if 'id' in col.lower()]
numerical_cols = [col for col in numerical_cols if col not in id_cols]

if not numerical_cols:
    print("No numerical columns found in features (X) to scale.")
else:
    print(f"Numerical columns in X to scale: {numerical_cols}")
    scaler = StandardScaler()
    X[numerical_cols] = scaler.fit_transform(X[numerical_cols])
    print("Numerical features in X scaled successfully.")

print("\nFeatures (X) after numerical scaling:")
```

Split Data into Training and Testing Sets

Finally, I'll split the preprocessed data into features (X) and a target variable (y), and then divide them into training and testing sets. For this example, I'll assume the target variable is named `target` or `risk`. If your target column has a different name, please modify the `target_column_name` variable below.

```
from sklearn.model_selection import train_test_split

# Split the preprocessed features (X) and target (y) into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"Data split into training and testing sets.")
print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"y_test shape: {y_test.shape}")

print("\nFirst 5 rows of X_train:")
display(X_train.head())
print("\nFirst 5 rows of y_train:")
display(y_train.head())
```

OUTPUT:

```
Archive:  /content/HealtCare_Risk_factor.zip
  inflating: /content/dirty_v3_path.csv
total 4620
-rw-r--r-- 1 root root 3379281 Sep 27  2025 dirty_v3_path.csv
-rw-r--r-- 1 root root 1342385 Apr  1 16:03 HealtCare_Risk_factor.zip
drwxr-xr-x 1 root root    4096 Mar 30 13:29 sample_data
```

First 5 rows of the dataset:

	Age	Gender	Medical Condition	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Smoking	Alcohol	Physical Activity	Diet Score	Family History	Stress Level	Sleep Hours	random_notes	noise_col
0	46.0	Male	Diabetes	137.04	135.27	28.90	96.04	6	231.88	210.56	7.61	0	0	-0.20	3.54	0	5.07	6.05	lorem	-137.057211
1	22.0	Male	Healthy	71.58	113.27	26.29	97.54	2	165.57	129.41	4.91	0	0	8.12	5.90	0	5.87	7.72	ipsum	-11.230610
2	50.0	NaN	Asthma	95.24	NaN	22.53	90.31	2	214.94	165.35	5.60	0	0	5.01	4.65	1	3.09	4.82	ipsum	98.331195
3	57.0	NaN	Obesity	NaN	130.53	38.47	96.60	5	197.71	182.13	6.92	0	0	3.16	3.37	0	3.01	5.33	lorem	44.187175
4	66.0	Female	Hypertension	95.15	178.17	31.12	94.90	4	259.53	115.85	5.98	0	1	3.56	3.40	0	6.38	6.64	lorem	44.831426

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 20 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Age                 25500 non-null  float64
 1   Gender              25500 non-null  object
 2   Medical Condition   25500 non-null  object
 3   Glucose             25500 non-null  float64
 4   Blood Pressure      25500 non-null  float64
 5   BMI                 30000 non-null  float64
 6   Oxygen Saturation   30000 non-null  float64
 7   LengthOfStay        30000 non-null  int64
 8   Cholesterol         30000 non-null  float64
 9   Triglycerides       30000 non-null  float64
 10  HbA1c               30000 non-null  float64
 11  Smoking             30000 non-null  int64
 12  Alcohol             30000 non-null  int64
 13  Physical Activity   30000 non-null  float64
 14  Diet Score          30000 non-null  float64
 15  Family History      30000 non-null  int64
 16  Stress Level        30000 non-null  float64
 17  Sleep Hours         30000 non-null  float64
 18  random_notes        30000 non-null  object
 19  noise_col           30000 non-null  float64
dtypes: float64(13), int64(4), object(3)
```

```
Missing values in 'Blood Pressure' filled with median: 138.32
No missing values in 'Cholesterol'.

Imputing other missing values:
  Missing values in numerical column 'Age' filled with median: 55.0
  Missing values in categorical column 'Gender' filled with mode: Female
  Missing values in categorical column 'Medical Condition' filled with mode: Hypertension
  Missing values in numerical column 'Glucose' filled with median: 110.5

Missing values after imputation:
```

	0
Age	0
Gender	0
Medical Condition	0
Glucose	0
Blood Pressure	0
BMI	0
Oxygen Saturation	0
LengthOfStay	0
Cholesterol	0
Triglycerides	0
HbA1c	0
Smoking	0
Alcohol	0
Physical Activity	0
Diet Score	0
Family History	0
Stress Level	0
Sleep Hours	0
random_notes	0
noise_col	0

dtype: int64

```
Target variable 'Medical Condition' separated successfully.
X shape: (30000, 19)
y shape: (30000,)
```

```
Categorical columns identified in features (X): ['Gender', 'random_notes']
Categorical columns in X encoded successfully.

Features (X) after one-hot encoding:
```

	Age	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Smoking	...	Family History	Stress Level	Sleep Hours	noise_col	Gender_Female	Gender_Male	random_notes_###	random_notes_??	random_notes_ipsum	random_notes_lorem
0	46.0	137.04	135.27	28.90	96.04	6	231.88	210.56	7.61	0	...	0	5.07	6.05	-137.057211	0.0	1.0	0.0	0.0	0.0	1.0
1	22.0	71.58	113.27	26.29	97.54	2	165.57	129.41	4.91	0	...	0	5.87	7.72	-11.230610	0.0	1.0	0.0	0.0	1.0	0.0
2	50.0	95.24	138.32	22.53	90.31	2	214.94	165.35	5.60	0	...	1	3.09	4.82	98.331195	1.0	0.0	0.0	0.0	1.0	0.0
3	57.0	110.50	130.53	38.47	96.60	5	197.71	182.13	6.92	0	...	0	3.01	5.33	44.187175	1.0	0.0	0.0	0.0	0.0	1.0
4	66.0	95.15	178.17	31.12	94.90	4	259.53	115.85	5.98	0	...	0	6.38	6.64	44.831426	1.0	0.0	0.0	0.0	0.0	1.0

5 rows × 23 columns

```
Numerical columns in X to scale: ['Age', 'Glucose', 'Blood Pressure', 'BMI', 'Oxygen Saturation', 'LengthOfStay', 'Cholesterol', 'HbA1c', 'Smoking', 'Alcohol', 'Physical Activity', 'Diet Score', 'Family History', 'Stress Level', 'Sleep Hours', 'noise_col', 'Gender_Female',
Numerical features in X scaled successfully.

Features (X) after numerical scaling:
```

	Age	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Smoking	...	Family History	Stress Level	Sleep Hours	noise_col	Gender_Female	Gender_Male	random_notes_###	random_notes_??	random_notes_ipsum	random_notes_lorem
0	-0.588993	0.398433	-0.250163	0.073986	0.290409	0.574279	0.562248	210.56	0.996497	-0.623249	...	-0.885386	-0.380811	-0.151421	-1.364375	-1.172330	1.172330	-0.581199	-0.579762	-0.576119	1.747272
1	-2.218620	-1.296692	-1.381416	-0.381614	0.691893	-0.874214	-1.416021	129.41	-1.048574	-0.623249	...	-0.885386	-0.021263	1.254717	-0.107055	-1.172330	1.172330	-0.581199	-0.579762	1.735754	-0.573321
2	-0.317388	-0.684002	-0.093330	-1.037957	-1.243260	-0.874214	0.056866	165.35	-0.525945	-0.623249	...	1.129450	-1.270691	-1.187081	0.987738	0.853002	-0.853002	-0.581199	-0.579762	1.735754	-0.573321
3	0.157920	-0.288836	-0.493897	1.744520	0.440296	0.212156	-0.457168	182.13	0.473868	-0.623249	...	-0.885386	-1.306645	-0.757661	0.446706	0.853002	-0.853002	-0.581199	-0.579762	-0.576119	1.747272
4	0.769030	-0.686333	1.955780	0.461508	-0.014719	-0.149967	1.387149	115.85	-0.238120	-0.623249	...	-0.885386	0.207948	0.345358	0.453143	0.853002	-0.853002	-0.581199	-0.579762	-0.576119	1.747272

5 rows × 23 columns

```
Data split into training and testing sets.
X_train shape: (24000, 23)
X_test shape: (6000, 23)
y_train shape: (24000,)
y_test shape: (6000,)
```

First 5 rows of X_train:

	Age	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Smoking	...	Family History	Stress Level	Sleep Hours	noise_col	Gender_Female	Gender_Male	random_notes_###	random_notes_??	random_notes_ipsum	random_notes_lorem
21753	-0.656894	-0.412358	0.771564	-0.929731	0.453679	-0.149967	0.569707	222.41	0.102725	-0.623249	...	1.129450	0.248397	-1.675440	1.860277	0.853002	-0.853002	-0.581199	-0.579762	-0.576119	1.747272
251	0.429524	2.552429	0.780820	-1.416751	0.584830	-0.149967	0.045231	186.82	1.231302	-0.623249	...	1.129450	-0.731370	-2.239580	0.983520	0.853002	-0.853002	-0.581199	-0.579762	-0.576119	1.747272
22941	-1.675411	-0.741232	-0.093330	1.051519	0.633008	-0.149967	-0.298751	131.86	-0.631985	-0.623249	...	-0.885386	0.095589	0.858978	-0.726465	-1.172330	1.172330	-0.581199	1.724845	-0.576119	-0.572321
618	1.515943	-0.505064	-0.093330	-1.168877	0.389441	0.212156	-0.588137	165.96	-0.419904	1.604496	...	-0.885386	-0.165082	-1.237601	-0.331466	0.853002	-0.853002	-0.581199	-0.579762	-0.576119	1.747272
17090	1.855448	-0.706273	0.646612	0.218870	-0.622298	0.212156	0.418152	277.23	0.125448	-0.623249	...	-0.885386	0.284352	0.707418	0.054187	-1.172330	1.172330	-0.581199	-0.579762	-0.576119	1.747272

5 rows × 23 columns

First 5 rows of y_train:

	Medical Condition
21753	Hypertension
251	Diabetes
22941	Obesity
618	Arthritis
17090	Hypertension

dtype: object

EXPLANATION:

- The HealtCare_Risk_factor.zip file was unzipped, and the dirty_v3_path.csv dataset was loaded into a DataFrame after correcting the filename.
- Missing values in numerical columns (like 'Age', 'Glucose', 'Blood Pressure') were filled with their median, and categorical columns ('Gender', 'Medical Condition') were filled with their mode.
- The target variable, 'Medical Condition', was separated from the features. Categorical features (Gender, random_notes) were then one-hot encoded, and all numerical features were scaled using StandardScaler.
- Finally, the preprocessed features and target variable were split into training and testing sets, making the data ready for model training.

Task 4 – Employee Salary Data Preprocessing

PROMPT:

/content/Employee_salary.zip Generate Python code to clean and preprocess an employee salary dataset for HR analytics. Handle missing values appropriately , Detect and remove salary outliers using a suitable method Standardize department names by converting them to lowercase and Apply label encoding to job_location and department columns Display the first few rows of the processed dataset and ensure it is ready for analysis

K /content/Employee_salary.zip Generate Python code to clean and preprocess an employee salary dataset for HR analytics. Handle missing values appropriately , Detect and remove salary outliers using a suitable method Standardize department names by converting them to lowercase and Apply label encoding to job_location and department columns Display the first few rows of the processed dataset and ensure it is ready for analysis

CODE:

12]
0s

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
```

Load the Employee Salary Dataset

First, we'll load the `employee_salary.csv` file into a pandas DataFrame. I'm assuming the file is located in the `/content/` directory. If your file is named differently or in another location, please modify the path accordingly.

13]
0s

```
import zipfile
import os

zip_file_path = '/content/Employee_salary.zip'
extraction_path = '/content/'

# Unzip the file
with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    zip_ref.extractall(extraction_path)

print(f"'{zip_file_path}' unzipped to '{extraction_path}'")
# Verify that the CSV file now exists
if 'employee_salary.csv' in os.listdir(extraction_path):
    print("employee_salary.csv found after unzipping.")
else:
    print("Warning: employee_salary.csv not found directly after unzipping. Please
```

```python
# Load the dataset
try:
    df = pd.read_csv('/content/Employee_Salary_Dataset.csv')
    print("Dataset loaded successfully.")
    display(df.head())
    display(df.info())
except FileNotFoundError:
    print("Error: 'Employee_Salary_Dataset.csv' not found. Please ensure the file i
    df = pd.DataFrame() # Create an empty DataFrame to avoid errors in subsequent s
```

bias.

[15]
✓ 0s

```python
# Identify numerical and categorical columns
if not df.empty:
    numerical_cols = df.select_dtypes(include=['number']).columns
    categorical_cols = df.select_dtypes(include=['object']).columns

    # Fill missing numerical values with the median
    for col in numerical_cols:
        if df[col].isnull().any():
            median_val = df[col].median()
            df[col].fillna(median_val, inplace=True)
            print(f"Missing values in '{col}' filled with median: {median_val}")

    # Fill missing categorical values with the mode
    for col in categorical_cols:
        if df[col].isnull().any():
            mode_val = df[col].mode()[0]
            df[col].fillna(mode_val, inplace=True)
            print(f"Missing values in '{col}' filled with mode: {mode_val}")

    print("\nMissing values after handling:")
    display(df.isnull().sum())
```

Detect and Remove Salary Outliers

We'll use the Interquartile Range (IQR) method to detect and remove outliers in the 'salary' column. This method is robust to skewed data.

[16]
✓ 0s

```python
if not df.empty and 'salary' in df.columns:
    Q1 = df['salary'].quantile(0.25)
    Q3 = df['salary'].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers_df = df[(df['salary'] < lower_bound) | (df['salary'] > upper_bound)]
    df_no_outliers = df[(df['salary'] >= lower_bound) & (df['salary'] <= upper_bound)]

    print(f"Number of salary outliers detected: {len(outliers_df)}")
    print(f"Original dataset size: {len(df)}")
    print(f"Dataset size after removing outliers: {len(df_no_outliers)}")

    df = df_no_outliers.copy() # Update the DataFrame to the version without outliers
else:
    print("Skipping outlier detection: 'salary' column not found or DataFrame is empty.")
```

Standardize Department Names

Converting department names to lowercase ensures consistency and avoids issues due to variations in capitalization (e.g., 'HR' vs 'hr').

```
if not df.empty and 'department' in df.columns and df['department'].dtype == 'object':
    df['department'] = df['department'].str.lower()
    print("Department names standardized to lowercase.")
    display(df['department'].value_counts().head())
else:
    print("Skipping department standardization: 'department' column not found or is not of object type, or DataFrame is empty.")
```

```
if not df.empty:
    le = LabelEncoder()
    columns_to_encode = ['job_location', 'department']

    for col in columns_to_encode:
        if col in df.columns and df[col].dtype == 'object':
            df[f'{col}_encoded'] = le.fit_transform(df[col])
            print(f"Label encoding applied to '{col}'.")
            # Optionally display mapping for verification
            # print(dict(zip(le.classes_, le.transform(le.classes_))))
        else:
            print(f"Skipping encoding: '{col}' column not found or is not of object type.")
else:
    print("Skipping label encoding: DataFrame is empty.")
```

Display Processed Dataset

Here are the first few rows of the cleaned and preprocessed dataset, ready for HR analytics.

```
if not df.empty:
    display(df.head())
    display(df.info())
else:
    print("DataFrame is empty after processing. Please check previous steps.")
```

OUTPUT:

```
'/content/Employee_salary.zip' unzipped to '/content/'
Warning: employee_salary.csv not found directly after unzipping. Please check the contents of the zip file.
```

```
Dataset loaded successfully.
```

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 5 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                35 non-null     int64
 1   Experience_Years  35 non-null     int64
 2   Age               35 non-null     int64
 3   Gender            35 non-null     object
 4   Salary            35 non-null     int64
dtypes: int64(4), object(1)
memory usage: 1.5+ KB
None
```

```
Skipping outlier detection: 'salary' column not found or DataFrame is empty.
```

```
Skipping department standardization: 'department' column not found or is not of object type, or DataFrame is empty.
```

```
Missing values after handling:
```

	0
ID	0
Experience_Years	0
Age	0
Gender	0
Salary	0

dtype: int64

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000

```
<class 'pandas.core.frame.DataFrame'>
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 0   ID                35 non-null     int64
 1   Experience_Years  35 non-null     int64
 2   Age               35 non-null     int64
 3   Gender            35 non-null     object
 4   Salary            35 non-null     int64
dtypes: int64(4), object(1)
memory usage: 1.5+ KB
None
```

EXPLANATION:

- The process started with loading the employee salary dataset into a pandas DataFrame and displaying its basic information.
- Numerical columns were intended to have missing values filled with their median, while categorical columns were to be filled with their mode.
- The Interquartile Range (IQR) method was planned to identify and remove outliers specifically from the 'salary' column.
- This included standardizing department names to lowercase for consistency and applying label encoding to categorical columns like 'job_location' and 'department' to prepare them for machine learning models.